# Density Functional Theory: From the Hamiltonian to Thermodynamic Properties

Alastair J. Price
*Guest Lecturer*

CHM328H1 S – Modern Physical Chemistry
University of Toronto – Winter 2026

# Lecture Overview

**Part I: Electronic Structure**

1. The Molecular Hamiltonian
2. Born–Oppenheimer Approximation
3. Hartree–Fock Theory
4. Density Functional Theory
5. Jacob's Ladder of Functionals

**Part II: From Energy to Thermodynamics**

6. The Potential Energy Surface
7. The Harmonic Approximation
8. Normal Modes & the Hessian
9. Partition Functions from DFT
10. Thermodynamic Properties

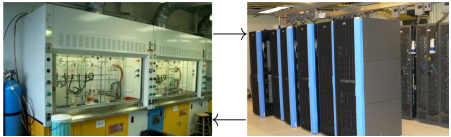
*Goal: Understand how a quantum-mechanical calculation on a computer gives you $\Delta G$, $\Delta H$, $K_{eq}$, and everything you learned in statistical thermodynamics.*

# Why Computational Chemistry?

**Theory and experiment are partners:**

- Predict properties **before** synthesizing
- Interpret experimental spectra (IR, NMR, UV-Vis)
- Access quantities that are hard to measure:
  - Transition state geometries
  - Reaction barrier heights
  - Individual energy contributions
- Guide rational molecular design

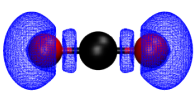
*Recall from Week 1: thermodynamics is empirical, not predictive. Computational chemistry fills that gap — we compute $\Delta G$ from first principles, no experiment needed.*

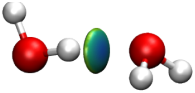


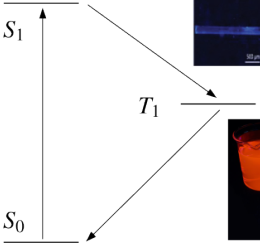
# What Can Theory Predict?

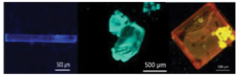
From a single calculation we can obtain:

- **Energies** – total, relative, binding
- **Structures** – bond lengths, angles, dihedrals
- **Vibrational frequencies** – IR/Raman spectra
- **Thermodynamic quantities** – $\Delta H$, $\Delta G$, $\Delta S$
- **Reaction profiles** – barriers, intermediates
- **Electronic properties** – charges, dipoles, orbitals
- **Spectroscopic parameters** – NMR shifts, excitation energies







# The Central Problem of Quantum Chemistry

**The Time-Independent Schrödinger Equation**

$$\hat{H}\Psi = E\Psi$$

- $\hat{H}$ = the Hamiltonian operator (total energy operator)
- $\Psi$ = the wavefunction (contains all information about the system)
- $E$ = the energy eigenvalue

**The challenge:** For any system with more than one electron, we **cannot solve this exactly**.

We need **approximations** – and different approximations give rise to different computational methods (HF, DFT, MP2, CCSD(T), ...).

# The Full Molecular Hamiltonian

*You've seen* $\hat{H} = \hat{T} + \hat{V} = \frac{1}{2}\sum \frac{p_I^2}{m_I} + V(\{q_i\})$ *since Week 1. Now we write the full* **molecular** *version:*

For a molecule with $M$ nuclei and $N$ electrons:

$$\hat{H} = \underbrace{-\sum_{A=1}^{M} \frac{\hbar^2}{2M_A}\nabla_A^2}_{\hat{T}_{\text{nuc}}} \underbrace{-\sum_{i=1}^{N} \frac{\hbar^2}{2m_e}\nabla_i^2}_{\hat{T}_{\text{elec}}} + \underbrace{\sum_{A>B}^{M} \frac{Z_A Z_B e^2}{r_{AB}}}_{V_{\text{nuc-nuc}}} \underbrace{-\sum_{i=1}^{N}\sum_{A=1}^{M} \frac{Z_A e^2}{r_{iA}}}_{V_{\text{elec-nuc}}} + \underbrace{\sum_{i>j}^{N} \frac{e^2}{r_{ij}}}_{V_{\text{elec-elec}}}$$

- $\hat{T}_{\text{nuc}}$: nuclear kinetic energy
- $\hat{T}_{\text{elec}}$: electron kinetic energy
- $V_{\text{nuc-nuc}}$: nuclear–nuclear repulsion
- $V_{\text{elec-nuc}}$: electron–nuclear attraction
- $V_{\text{elec-elec}}$: electron–electron repulsion ← **the hard part!**

# Atomic Units – Simplifying the Notation

In **atomic units** (a.u.), we set $\hbar = m_e = e = 4\pi\epsilon_0 = 1$:

$$\hat{H} = -\sum_{A=1}^{M} \frac{1}{2M_A}\nabla_A^2 - \frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 + \sum_{A>B}^{M} \frac{Z_A Z_B}{R_{AB}} - \sum_{i=1}^{N}\sum_{A=1}^{M} \frac{Z_A}{r_{iA}} + \sum_{i>j}^{N} \frac{1}{r_{ij}}$$

| Quantity | Atomic Unit | SI Equivalent |
|---|---|---|
| Length | 1 bohr ($a_0$) | $5.292 \times 10^{-11}$ m |
| Energy | 1 hartree ($E_h$) | $4.360 \times 10^{-18}$ J = 627.5 kcal/mol |
| Mass | $m_e$ | $9.109 \times 10^{-31}$ kg |
| Charge | $e$ | $1.602 \times 10^{-19}$ C |

# Example: The $H_2^+$ Molecular Ion

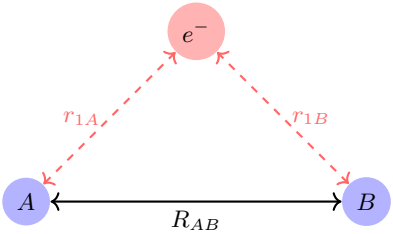
The simplest molecule: 2 nuclei (A, B), 1 electron.

**Full Hamiltonian** (atomic units):

$$\hat{H} = -\frac{1}{2M_A}\nabla_A^2 - \frac{1}{2M_B}\nabla_B^2 - \frac{1}{2}\nabla_1^2 - \frac{1}{r_{1A}} - \frac{1}{r_{1B}} + \frac{1}{R_{AB}}$$

**Key point:** Even for this "simple" system, the nuclear and electronic coordinates are coupled through $r_{1A}$ and $r_{1B}$.

→ We need the **Born–Oppenheimer approximation** to make progress.




*From your homework 2, Problem 1!*

# The Born–Oppenheimer Approximation

**Key Idea**

Nuclei are $\sim 1836\times$ heavier than electrons $\Rightarrow$ electrons respond **instantaneously** to nuclear motion.

**Separation of variables:**

$$\Psi_{\text{total}}(\mathbf{r}, \mathbf{R}) \approx \psi_{\text{elec}}(\mathbf{r}; \mathbf{R}) \cdot \chi_{\text{nuc}}(\mathbf{R})$$

1. **Fix** the nuclear positions $\mathbf{R}$ (they become parameters, not variables)
2. **Solve** the electronic Schrödinger equation:

$$\hat{H}_{\text{elec}}\, \psi_{\text{elec}}(\mathbf{r}; \mathbf{R}) = E_{\text{elec}}(\mathbf{R})\, \psi_{\text{elec}}(\mathbf{r}; \mathbf{R})$$

3. The electronic energy $E_{\text{elec}}(\mathbf{R})$ **depends on nuclear positions** $\rightarrow$ this creates the **Potential Energy Surface (PES)**
4. Nuclei move on this PES

# The Electronic Hamiltonian

After the Born–Oppenheimer approximation, we solve:

$$\hat{H}_{\text{elec}} = -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{Z_A}{r_{iA}} + \sum_{i>j}^{N}\frac{1}{r_{ij}}$$

- Nuclear–nuclear repulsion $V_{nn} = \sum_{A>B}\frac{Z_A Z_B}{R_{AB}}$ is just a **constant** for fixed $\mathbf{R}$
- The total energy at a given geometry is:

$$E_{\text{total}}(\mathbf{R}) = E_{\text{elec}}(\mathbf{R}) + V_{nn}(\mathbf{R})$$

**The Electronic Structure Problem**

Given: nuclear positions $\mathbf{R}$ and number of electrons $N$
Find: $E_{\text{elec}}(\mathbf{R})$ and $\psi_{\text{elec}}(\mathbf{r};\mathbf{R})$
**This is what DFT (and HF, MP2, CC, ...) solves!**

# Approach 1: Wavefunction-Based Methods

**Idea:** Approximate the many-electron wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ directly.

**Hartree–Fock (HF) Theory:**

- Approximate $\Psi$ as a single **Slater determinant** (antisymmetrized product of one-electron orbitals):

$$\Psi_{\text{HF}} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\mathbf{r}_1) & \phi_2(\mathbf{r}_1) & \cdots & \phi_N(\mathbf{r}_1) \\ \phi_1(\mathbf{r}_2) & \phi_2(\mathbf{r}_2) & \cdots & \phi_N(\mathbf{r}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{r}_N) & \phi_2(\mathbf{r}_N) & \cdots & \phi_N(\mathbf{r}_N) \end{vmatrix}$$

- Automatically satisfies the **Pauli exclusion principle** (antisymmetry)
- Each electron moves in the **average field** of all other electrons

# Hartree–Fock Energy Expression

The HF energy is obtained by minimizing $\left\langle \Psi_{\text{HF}} | \hat{H} | \Psi_{\text{HF}} \middle| \Psi_{\text{HF}} | \hat{H} | \Psi_{\text{HF}} \right\rangle$:

$$E_{\text{HF}} = \sum_i h_i + \frac{1}{2} \sum_{ij} (J_{ij} - K_{ij})$$

- $h_i$ = one-electron integrals
  (kinetic energy + nuclear attraction)
- $J_{ij}$ = **Coulomb integrals**
  (classical electron–electron repulsion)
- $K_{ij}$ = **Exchange integrals**
  (quantum mechanical, no classical analogue)
- **Missing:** electron **correlation**
  (electrons avoid each other beyond mean-field)

**Correlation energy:** $E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}$
Typically small ($\sim$1% of total energy) but **chemically crucial** ($\sim$1 kcal/mol = make or break for chemistry!)

# What is Exchange?

**Exchange** is a purely **quantum mechanical** effect arising from the antisymmetry of the wavefunction (Pauli exclusion principle).

**Physical picture:**

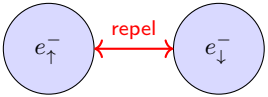
- Two electrons with the **same spin** cannot be at the same point in space
- This creates an "exchange hole" – a region of depleted probability around each electron
- Electrons with parallel spins are kept apart $\Rightarrow$ their repulsion is **reduced**

**In the HF energy:**

$$K_{ij} = \iint \frac{\phi_i^*(\mathbf{r}_1)\phi_j(\mathbf{r}_1)\,\phi_j^*(\mathbf{r}_2)\phi_i(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}\,d\mathbf{r}_1\,d\mathbf{r}_2$$

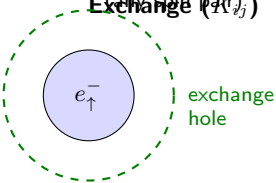
- $K_{ij} > 0$ always $\Rightarrow$ exchange **lowers** the energy ($-K_{ij}$ in $E_{\text{HF}}$)
- Non-zero only for same-spin electron pairs
- No classical analogue – purely quantum!

**Coulomb ($J_{ij}$)**




Classical repulsion

**Exchange ($K_{ij}$)** (any spin pair)




Same-spin electrons kept apart

# What is Electron Correlation?

**Correlation** is everything that HF **misses** about electron–electron interactions.

**The HF approximation:**

- Each electron moves in the **average** field of all others
- Electrons don't "see" each other's instantaneous positions
- Like driving on a highway knowing only the average traffic density, not where each car actually is

**Reality:**

- Electrons **avoid each other** dynamically (they are correlated)
- This creates a "Coulomb hole" around each electron
- Correlation **always lowers** the energy: $E_{\text{corr}} < 0$

## Types of Correlation

**Dynamic correlation:**
Short-range avoidance of electrons. Smooth, small effect per pair, but adds up. Captured by MP2, CCSD.

**Static (non-dynamic):**
Near-degeneracy effects. Multiple configurations contribute equally. Needs multi-reference methods (CASSCF).

**Key numbers:**
$E_{\text{corr}}(\text{H}_2\text{O}) \approx -0.37\ E_h$
$\approx$ 1% of total energy
$\approx$ 230 kcal/mol!

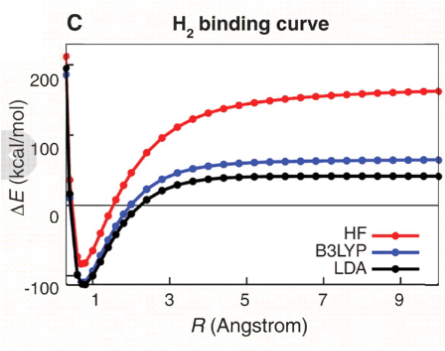
# The Electron Correlation Problem

**Post-HF methods** recover correlation:

- **MP2**: 2nd-order perturbation theory
- **CCSD**: coupled cluster singles & doubles
- **CCSD(T)**: "gold standard" of quantum chemistry
- **Full CI**: exact (within basis set), but scales as $\sim N!$

**Problem:** Cost scales steeply with system size!

- HF: $\mathcal{O}(N^3\text{–}N^4)$
- MP2: $\mathcal{O}(N^5)$
- CCSD: $\mathcal{O}(N^6)$
- CCSD(T): $\mathcal{O}(N^7)$



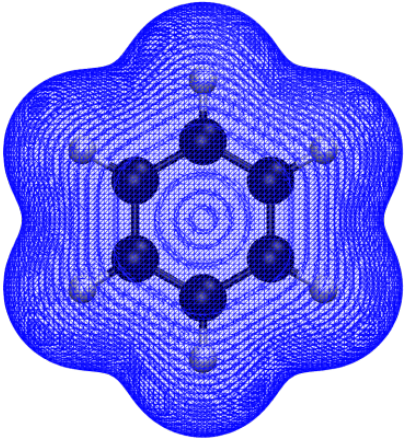

$\Rightarrow$ Need a method that includes correlation at **lower cost**... **Enter DFT!**

# Approach 2: Density Functional Theory

**Key insight:** Instead of the wavefunction $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ (depends on $3N$ coordinates), use the **electron density** $\rho(\mathbf{r})$ (depends on only 3 coordinates!).

$$\rho(\mathbf{r}) = N \int |\Psi(\mathbf{r}, \mathbf{r}_2, \ldots, \mathbf{r}_N)|^2 \, d\mathbf{r}_2 \cdots d\mathbf{r}_N$$

- $\rho(\mathbf{r}) \geq 0$ everywhere
- $\int \rho(\mathbf{r}) \, d\mathbf{r} = N$
- $\rho$ is a **physical observable** (measurable by X-ray diffraction!)



Electron density of benzene

# Hohenberg–Kohn Theorems (1964)

## Theorem 1: Existence

The ground-state electron density $\rho_0(\mathbf{r})$ **uniquely determines** the external potential $v_{\text{ext}}(\mathbf{r})$ (and hence all ground-state properties).

$$\rho_0(\mathbf{r}) \xrightarrow{\text{1-to-1}} v_{\text{ext}}(\mathbf{r}) \to \hat{H} \to \Psi_0 \to E_0, \text{all properties}$$

## Theorem 2: Variational Principle

The ground-state energy is a **functional** of the density, and:

$$E[\rho] \geq E_0 \quad \text{for all trial densities } \rho$$

The true ground-state density minimizes the energy functional.

**Great news:** Everything is determined by $\rho(\mathbf{r})$.
**Bad news:** We don't know the exact form of $E[\rho]$!

# Kohn–Sham DFT (1965)

**Kohn & Sham's brilliant idea:** Introduce a **fictitious non-interacting system** with the same density as the real one.

The Kohn–Sham energy functional:

$$E_{\mathsf{KS}}[\rho] = \underbrace{T_s[\rho]}_{\text{KS kinetic}} + \underbrace{\int v_{\mathsf{ext}}(\mathbf{r})\rho(\mathbf{r})\,d\mathbf{r}}_{E_{\mathsf{ext}}} + \underbrace{\frac{1}{2}\iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d\mathbf{r}\,d\mathbf{r}'}_{J[\rho]\ \text{(Coulomb)}} + \underbrace{E_{\mathsf{XC}}[\rho]}_{\text{exchange-correlation}}$$

- $T_s[\rho]$: kinetic energy of **non-interacting** KS electrons (exact for KS orbitals)
- $J[\rho]$: classical Coulomb repulsion
- $E_{\mathsf{XC}}[\rho]$: **everything else** – exchange, correlation, correction to $T_s$

**The XC functional $E_{\mathsf{XC}}[\rho]$ is the only unknown!**
All of DFT development is about finding better approximations to $E_{\mathsf{XC}}$.

# The Kohn–Sham Equations

Minimizing $E_{\mathsf{KS}}[\rho]$ leads to one-electron equations:

$$\left[-\frac{1}{2}\nabla^2 + v_{\mathsf{eff}}(\mathbf{r})\right]\phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r})$$

where the effective potential is:

$$v_{\mathsf{eff}}(\mathbf{r}) = v_{\mathsf{ext}}(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}' + v_{\mathsf{XC}}(\mathbf{r})$$

**Self-Consistent Field (SCF) procedure:**

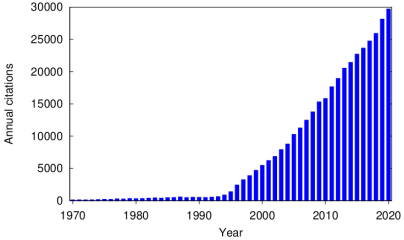
1. Guess initial density $\rho^{(0)}(\mathbf{r})$
2. Construct $v_{\mathsf{eff}}$ from current $\rho$
3. Solve KS equations $\rightarrow$ get orbitals $\{\phi_i\}$
4. Compute new density: $\rho^{(n+1)}(\mathbf{r}) = \sum_{i=1}^{N} |\phi_i(\mathbf{r})|^2$
5. If converged, stop. Otherwise, go to step 2.

# HF vs. DFT – A Comparison

| | **Hartree–Fock** | **Kohn–Sham DFT** |
|---|---|---|
| Central quantity | Wavefunction $\Psi$ | Density $\rho(\mathbf{r})$ |
| Equations | HF equations | KS equations |
| Exchange | Exact (from $\Psi$) | Approximated in $E_{\text{XC}}$ |
| Correlation | **Missing** | Included in $E_{\text{XC}}$ |
| Cost | $\mathcal{O}(N^3\text{–}N^4)$ | $\mathcal{O}(N^3)$ |
| Systematic improvement | Yes (MP2, CC, ...) | **No** (no "Jacob's ladder" guarantee) |

**DFT advantage:** Includes correlation at nearly the same cost as HF!
**DFT disadvantage:** The exact $E_{\text{XC}}$ is unknown – results depend on the chosen functional.

# Jacob's Ladder of Density Functional Approximations



Walter Kohn

Axel Becke

**Perdew's "Jacob's Ladder"** – increasing accuracy (and cost):

1. **LDA**: $E_{\mathrm{XC}}[\rho]$
   Local density only; uniform electron gas
2. **GGA**: $E_{\mathrm{XC}}[\rho, \nabla\rho]$
   PBE, BLYP – includes density gradient
3. **meta-GGA**: $E_{\mathrm{XC}}[\rho, \nabla\rho, \tau]$
   TPSS, SCAN – adds kinetic energy density
4. **Hybrid**: mix in exact (HF) exchange
   B3LYP, PBE0 – the workhorses of chemistry!
5. **Double hybrid**: + MP2 correlation
   Most accurate, most expensive

# Common Functionals in Chemistry

| Functional | Rung | HF Exchange | Common Use |
|---|---|---|---|
| LDA (SVWN) | 1 | 0% | Solids (outdated for molecules) |
| PBE | 2 (GGA) | 0% | Solids, surfaces |
| BLYP | 2 (GGA) | 0% | Organic chemistry |
| B3LYP | 4 (Hybrid) | 20% | **Most popular in chemistry** |
| PBE0 | 4 (Hybrid) | 25% | General purpose |
| $\omega$B97X-D | 4 (RSH) | varies | NCI, thermo |
| B2PLYP | 5 (DH) | 53% | High accuracy |

**Rules of thumb:**

- For organic reactions: **B3LYP-D3** or **$\omega$B97X-D** with a triple-$\zeta$ basis set
- For transition metals: **PBE0** or **TPSSh**
- Always consider **dispersion corrections** (D3, D4, XDM) for non-covalent interactions!

# Basis Sets – Expanding the Orbitals

KS orbitals $\phi_i(\mathbf{r})$ are expanded in a **basis set**:

$$\phi_i(\mathbf{r}) = \sum_{\mu} c_{\mu i}\, \chi_\mu(\mathbf{r})$$

where $\chi_\mu$ are basis functions centered on atoms (usually Gaussians).

**Common naming:** Pople or Dunning basis sets

- **STO-3G**: minimal (1 function per orbital) – **too small for production**
- **6-31G(d)**: split-valence + polarization – okay for quick checks
- **6-311+G(d,p)**: triple-$\zeta$ + diffuse + polarization
- **cc-pVDZ, cc-pVTZ, cc-pVQZ**: correlation-consistent, systematically improvable
- **def2-TZVP**: efficient, well-balanced for DFT

**Rule:** At minimum a **triple-$\zeta$** basis with **polarization** for reliable results.
Larger basis $\rightarrow$ better results, but more expensive.

# Decoding Basis Set Names

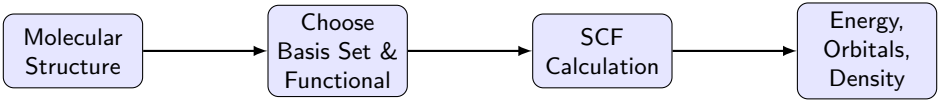
**Example:** What does **6-311+G(d,p)** actually mean?

- **6**: Core orbitals use a contraction of 6 primitive Gaussians
- **311**: Valence orbitals split into **three** sets (triple-$\zeta$)
  - 3 primitives (tight), 1 primitive (medium), 1 primitive (diffuse)
- **G**: Gaussian-type orbitals (not Slater-type)
- **+**: Diffuse functions on heavy atoms (for anions, weak interactions)
- **(d,p)**: Polarization functions — $d$ on heavy atoms, $p$ on H

**Other common families:**

| Name | Description |
|---|---|
| cc-pVDZ / cc-pVTZ / cc-pVQZ | Correlation-consistent: DZ, TZ, QZ |
| aug-cc-pVTZ | + diffuse functions (for anions, weak interactions) |
| def2-SVP / def2-TZVP | Karlsruhe: efficient, well-balanced for DFT |

# DFT Workflow in Practice

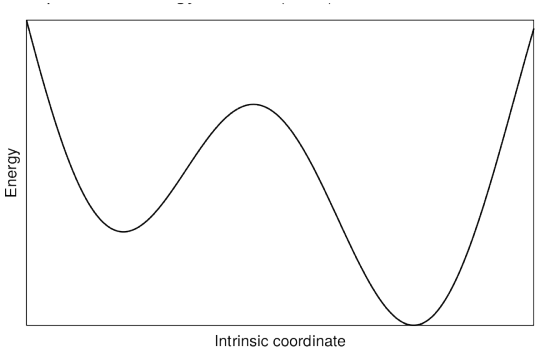



**Input requires:**

- Atomic coordinates (Cartesian or Z-matrix)
- Charge and spin multiplicity
- Choice of method (functional) and basis set
- Type of calculation (energy, optimization, frequency, ...)

**Software:** Gaussian, ORCA, Q-Chem, Psi4, FHI-aims, Quantum ESPRESSO, ...

# The Potential Energy Surface (PES)



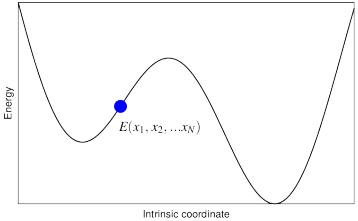

$E_{\text{total}}(\mathbf{R})$ as a function of all nuclear coordinates defines the **PES**.

For $M$ atoms: PES has $3M - 6$ dimensions ($3M - 5$ for linear).

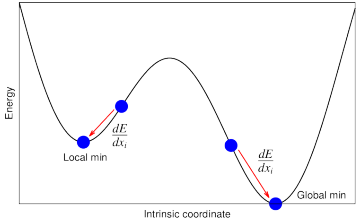
**Key features:**

- **Minima**: stable structures (reactants, products, intermediates)
- **Saddle points**: transition states
- **Gradient** $= \nabla E$: forces on atoms
- **Hessian** $= \nabla^2 E$: curvature (force constants)

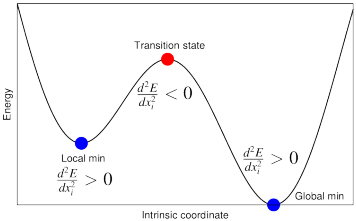
# Types of Calculations on the PES




**Single-Point Energy**
Energy at one fixed geometry.
$E(\mathbf{R}_0)$




**Geometry Optimization**
Follow gradient downhill to find minimum.
$\nabla E = \mathbf{0}$




**Frequency Calculation**
Compute Hessian $\rightarrow$ normal modes.
Verify min. or TS.

**At a minimum:** all frequencies real (positive eigenvalues of Hessian)
**At a transition state:** exactly **one imaginary frequency** (one negative eigenvalue)

# Geometry Optimization – Finding Minima

**Algorithm:** Follow the forces (negative gradient) downhill:

1. Start with initial geometry $\mathbf{R}_0$
2. Compute energy $E(\mathbf{R}_0)$ and gradient $\mathbf{g}_0 = \nabla E|_{\mathbf{R}_0}$
3. Take a step: $\mathbf{R}_1 = \mathbf{R}_0 - \alpha\,\mathbf{g}_0$ (steepest descent)
4. Repeat until $|\mathbf{g}| <$ threshold and $|\Delta E| <$ threshold

**Better algorithms:** Newton–Raphson uses Hessian information:

$$\mathbf{R}_{n+1} = \mathbf{R}_n - \mathbf{H}^{-1}\,\mathbf{g}_n$$

**Convergence criteria** (typical):

- Max force $< 4.5 \times 10^{-4}$ hartree/bohr
- RMS force $< 3.0 \times 10^{-4}$ hartree/bohr
- Max displacement $< 1.8 \times 10^{-3}$ bohr
- RMS displacement $< 1.2 \times 10^{-3}$ bohr

# Beyond Electronic Energy – We Need Vibrations!

**DFT gives us** $E_{\text{elec}}(\mathbf{R})$ – but thermodynamics requires knowledge of **nuclear motion**.

**The bridge:**

$$E_{\text{elec}}(\mathbf{R}) \xrightarrow{\text{harmonic approx.}} \text{frequencies } \{\nu_k\} \xrightarrow{\text{stat. mech.}} q_{\text{vib}} \to \Delta G, \Delta H, \Delta S, K_{\text{eq}}, k(T)$$

**The Key Connection**

The harmonic approximation is the **bridge** between
**quantum chemistry** (electronic structure)
and
**statistical thermodynamics** (macroscopic properties)

# The Morse Potential and Taylor Expansion

**True potential** for a diatomic (Morse):

$$V(r) = D_e \left(1 - \mathrm{e}^{-a(r-r_e)}\right)^2$$

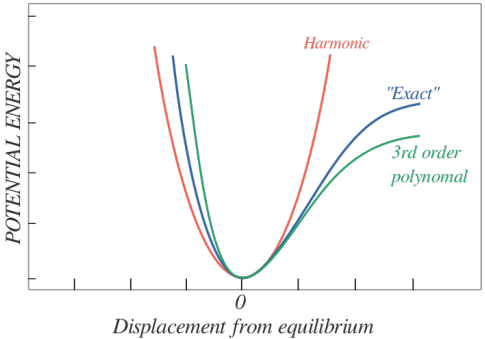
**Taylor expand** around equilibrium $r_e$:

$$V(r) \approx \cancel{V(r_e)} + \cancelto{0}{\left.\frac{dV}{dr}\right|_{r_e}} (r-r_e) + \frac{1}{2} \underbrace{\left.\frac{d^2V}{dr^2}\right|_{r_e}}_{k} (r-r_e)^2 + \cdots$$

$\Rightarrow$ Near equilibrium: $\boxed{V(r) \approx \frac{1}{2}k(r-r_e)^2}$

- $k = \left.\frac{d^2V}{dr^2}\right|_{r_e}$ is the **force constant**
- Harmonic approx. valid near minimum
- Breaks down far from $r_e$ (dissociation)

*From your homework 2, Problem 2!*




Comparison of harmonic (parabolic) approximation with the "exact" (Morse-like) potential.

# Force Constants and the Hessian Matrix

For a **polyatomic molecule**, the Taylor expansion generalizes to:

$$V(\mathbf{R}) \approx V(\mathbf{R}_e) + \frac{1}{2} \sum_{i,j} \underbrace{\left. \frac{\partial^2 V}{\partial R_i \partial R_j} \right|_{\mathbf{R}_e}}_{H_{ij}} \Delta R_i \, \Delta R_j$$

## The Hessian Matrix

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 V}{\partial R_1^2} & \frac{\partial^2 V}{\partial R_1 \partial R_2} & \cdots \\ \frac{\partial^2 V}{\partial R_2 \partial R_1} & \frac{\partial^2 V}{\partial R_2^2} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

The Hessian is a $3M \times 3M$ matrix of second derivatives of the energy with respect to nuclear displacements.

*Note: In your Week 4 lectures, Anatole called this* $\mathbf{M}$. *Same matrix, different letter!*

# Mass-Weighted Coordinates and Normal Modes

**Problem:** Cartesian displacements couple different atoms.

**Solution:** Transform to **mass-weighted coordinates** (exactly as in Week 4):

$$q_i = \sqrt{m_i}\,\Delta R_i$$

The mass-weighted Hessian is:

$$\tilde{H}_{ij} = \frac{H_{ij}}{\sqrt{m_i m_j}} = \frac{1}{\sqrt{m_i m_j}} \frac{\partial^2 V}{\partial R_i \partial R_j}$$

**Diagonalize $\tilde{\mathbf{H}}$:**

$$\tilde{\mathbf{H}}\,\mathbf{L} = \mathbf{L}\,\mathbf{\Lambda}$$

- **Eigenvalues** $\lambda_k$: related to vibrational frequencies $\nu_k = \frac{1}{2\pi}\sqrt{\lambda_k}$
- **Eigenvectors**: the **normal modes** – independent vibrational motions
- 6 eigenvalues $= 0$ (translations $+$ rotations); $3M - 6$ are vibrational

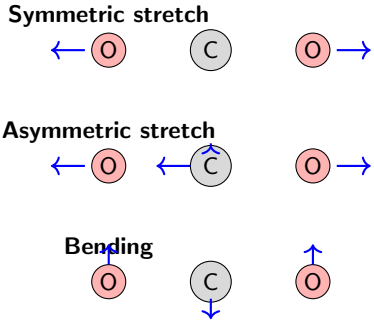
# Normal Modes – Physical Meaning

**Normal modes** are **independent** vibrations where all atoms move in phase at the same frequency.

**Example: Linear triatomic** ($CO_2$):

- $3(3) - 5 = 4$ vibrational modes
- Symmetric stretch: $\nu_1 = 1388$ cm$^{-1}$
- Asymmetric stretch: $\nu_3 = 2349$ cm$^{-1}$
- Bending (2× degenerate): $\nu_2 = 667$ cm$^{-1}$

*From your homework 2, Problem 3!*
You diagonalized the Hessian for a linear triatomic to find these modes.



# The Harmonic Oscillator – Quantum Energy Levels

Each normal mode $k$ behaves as an independent **quantum harmonic oscillator**:

$$\varepsilon_{k,n} = \left(n + \frac{1}{2}\right) h\nu_k = \left(n + \frac{1}{2}\right) \hbar\omega_k, \qquad n = 0, 1, 2, \ldots$$

*You derived $E = \hbar\omega(n + \frac{1}{2})$ in Week 4. Same result — $h\nu = \hbar\omega$ since $\omega = 2\pi\nu$.*

**Key features:**

- **Zero-point energy (ZPE):** Even at $T = 0$ K, $\varepsilon_{k,0} = \frac{1}{2}h\nu_k \neq 0$
- Total ZPE: $E_{\text{ZPE}} = \frac{1}{2}\sum_{k=1}^{3M-6} h\nu_k$
- Equally spaced energy levels (spacing $= h\nu_k$)
- Good approximation near the bottom of the potential well

**Important:** The **zero-point energy** is a quantum effect with no classical analogue. It affects reaction energies and must always be included!

## Statistical Mechanics Recap

**Connection:** Microscopic energy levels → macroscopic thermodynamics

**Canonical partition function** (using $\beta = 1/k_B T$ as in your lectures):

$$Q(N, V, T) = \sum_j \mathrm{e}^{-\beta E_j} = \sum_j \mathrm{e}^{-E_j/k_B T}$$

For $N$ **indistinguishable, independent** molecules:

$$Q = \frac{q^N}{N!}$$

where $q$ is the **molecular partition function**.

**Key thermodynamic relations** (you've derived these since Week 6):

$$A = -k_B T \ln Q \qquad \text{(Helmholtz free energy)}$$

$$U = \langle E \rangle = -\frac{\partial \ln Q}{\partial \beta} = k_B T^2 \left( \frac{\partial \ln Q}{\partial T} \right)_{N,V} \qquad \text{(internal energy)}$$

$$S = k_B \, \partial_T (T \ln Q) \qquad \text{(entropy)}$$

# Factorization of the Molecular Partition Function

Within the Born–Oppenheimer and rigid-rotor/harmonic-oscillator (RRHO) approximations:

$$\boxed{q = q_{\text{trans}} \cdot q_{\text{rot}} \cdot q_{\text{vib}} \cdot q_{\text{elec}}}$$

Each factor corresponds to an **independent** degree of freedom:

| **Factor** | **Information from DFT** | **Depends on** |
|---|---|---|
| $q_{\text{trans}}$ | Molecular mass | $T$, $V$ (or $P$) |
| $q_{\text{rot}}$ | Moments of inertia (from optimized geometry) | $T$ |
| $q_{\text{vib}}$ | Vibrational frequencies (from Hessian) | $T$ |
| $q_{\text{elec}}$ | Electronic energy, degeneracy | $T$ (usually $\approx 1$) |

**DFT frequency calculation gives us everything we need!**

# Translational Partition Function

*You derived this from PIB eigenvalues in Week 6:* $q^{trans} = (2\pi m/\beta h^2)^{3/2} V$.

For an ideal gas in a box of volume $V$:

$$q_{\text{trans}} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V = \frac{V}{\Lambda^3}$$

where $\Lambda$ is the **thermal de Broglie wavelength**:

$$\Lambda = \frac{h}{\sqrt{2\pi m k_B T}}$$

- $m =$ total molecular mass (sum of atomic masses)
- Typically $q_{\text{trans}} \sim 10^{30}$ at room temperature
- Dominant contribution to the entropy
- Often evaluated at standard pressure: $V = k_B T / P^\circ$

# Rotational Partition Function

For a **nonlinear polyatomic** molecule (rigid rotor, high-$T$ limit):

$$q_{\text{rot}} = \frac{\sqrt{\pi}}{\sigma}\left(\frac{8\pi^2 k_B T}{h^2}\right)^{3/2}\sqrt{I_A I_B I_C}$$

- $I_A, I_B, I_C$ = principal moments of inertia (from optimized geometry)
- $\sigma$ = **rotational symmetry number**
  - $H_2O$: $\sigma = 2$ ($C_{2v}$)
  - $NH_3$: $\sigma = 3$ ($C_{3v}$)
  - $CH_4$: $\sigma = 12$ ($T_d$)
  - $C_6H_6$: $\sigma = 12$ ($D_{6h}$)
- For a **linear molecule**: $q_{\text{rot}} = \frac{8\pi^2 I k_B T}{\sigma h^2}$

# Vibrational Partition Function

For a **harmonic oscillator** with frequency $\nu_k$:

$$q_{\text{vib},k} = \frac{1}{1 - \mathrm{e}^{-h\nu_k/k_B T}} = \frac{1}{1 - \mathrm{e}^{-\Theta_{V,k}/T}}$$

where $\Theta_{V,k} = h\nu_k/k_B$ is the **vibrational temperature**.

Total vibrational partition function (modes are independent):

$$q_{\text{vib}} = \prod_{k=1}^{3M-6} q_{\text{vib},k} = \prod_{k=1}^{3M-6} \frac{1}{1 - \mathrm{e}^{-\Theta_{V,k}/T}}$$

**Key limits:**

- $T \ll \Theta_V$: mode is "frozen", $q_{\text{vib},k} \approx 1$ (only ground state populated)
- $T \gg \Theta_V$: classical limit, $q_{\text{vib},k} \approx T/\Theta_V = k_B T/h\nu_k$

**Example:** C–H stretch $\sim 3000$ cm$^{-1}$ $\Rightarrow \Theta_V \approx 4300$ K (frozen at RT)
Low-frequency torsion $\sim 100$ cm$^{-1}$ $\Rightarrow \Theta_V \approx 144$ K (classical at RT)

# Vibrational Partition Function – Including ZPE

**Two conventions:**

**Bottom-of-the-well (BOW):**

$$q_{\text{vib}}^{\text{BOW}} = \prod_{k=1}^{3M-6} \frac{\mathrm{e}^{-\Theta_{V,k}/2T}}{1-\mathrm{e}^{-\Theta_{V,k}/T}}$$

Energy measured from the minimum of the PES.

**Zero-point exclusive (V=0):**

$$q_{\text{vib}}^{\text{V=0}} = \prod_{k=1}^{3M-6} \frac{1}{1-\mathrm{e}^{-\Theta_{V,k}/T}}$$

Energy measured from the ZPE level.

Most quantum chemistry programs (Gaussian, ORCA) use the BOW convention and report total energy as: $E_{\text{total}} = E_{\text{elec}} + E_{\text{ZPE}} + E_{\text{thermal}}$

# Electronic Partition Function

$$q_{\text{elec}} = \sum_j g_j \, \mathrm{e}^{-\varepsilon_j / k_B T}$$

For most closed-shell molecules at room temperature:

- Electronic excitation energies $\gg k_B T$ ($\sim$0.026 eV at 298 K)
- Only the ground state is populated
- $q_{\text{elec}} \approx g_0 = 1$ for singlet ground states

**Exceptions:**

- Open-shell species: $g_0 = 2S + 1$ (spin degeneracy)
- Example: $O_2$ ($^3\Sigma_g^-$): $g_0 = 3$
- NO ($^2\Pi$): low-lying excited states contribute at high $T$

# Putting It All Together: $G$ from DFT

**Total Gibbs free energy** from a DFT frequency calculation:

$$\boxed{G = E_{\text{elec}} + E_{\text{ZPE}} + H_{\text{thermal}} - TS}$$

where:

$$\begin{aligned} E_{\text{ZPE}} &= \frac{1}{2}\sum_k h\nu_k \\ H_{\text{thermal}} &= U_{\text{trans}} + U_{\text{rot}} + U_{\text{vib}} + k_B T \\ &= \frac{3}{2}k_B T + \frac{3}{2}k_B T + \sum_k \frac{h\nu_k}{\mathrm{e}^{h\nu_k/k_B T} - 1} + k_B T \\ S &= S_{\text{trans}} + S_{\text{rot}} + S_{\text{vib}} + S_{\text{elec}} \end{aligned}$$

**Key point:** $E_{\text{elec}}$ dominates ($\sim$thousands of hartree), but the **thermal corrections** ($\sim$millihartree) determine **relative** energies and selectivities!

# Entropy Contributions

**Translational entropy** (Sackur–Tetrode equation):

$$S_{\text{trans}} = R\left[\ln\left(\frac{q_{\text{trans}}}{N}\right) + \frac{5}{2}\right]$$

**Rotational entropy:**

$$S_{\text{rot}} = R\left[\ln q_{\text{rot}} + \frac{3}{2}\right] \quad \text{(nonlinear)}$$

**Vibrational entropy:**

$$S_{\text{vib}} = R\sum_k \left[\frac{\Theta_{V,k}/T}{\mathrm{e}^{\Theta_{V,k}/T} - 1} - \ln\left(1 - \mathrm{e}^{-\Theta_{V,k}/T}\right)\right]$$

- $S_{\text{trans}}$ is the **largest** contribution (typically ∼150 J/(mol·K))
- $S_{\text{vib}}$ matters for low-frequency modes ($\nu < 200$ cm$^{-1}$)
- At $T \to 0$: $S \to 0$ (third law of thermodynamics, ✓)

# What a Gaussian/ORCA Output Gives You

After a `freq` calculation, you get a thermochemistry summary:

```
Zero-point correction= 0.044749 (Hartree/Particle)
Thermal correction to Energy= 0.047545
Thermal correction to Enthalpy= 0.048489
Thermal correction to Gibbs Free Energy= 0.021178
Sum of electronic and zero-point Energies= -76.377939
Sum of electronic and thermal Energies= -76.375143
Sum of electronic and thermal Enthalpies= -76.374199
Sum of electronic and thermal Free Energies= -76.401510
```

- Use "Sum of electronic and thermal Free Energies" for $\Delta G$
- Compare **relative** values (products − reactants)
- Default: $T = 298.15$ K, $P = 1$ atm (can be changed)

# Computing Reaction Thermodynamics

**Example:** Reaction A $\longrightarrow$ B

$$\Delta E = E_{\text{elec}}(B) - E_{\text{elec}}(A) \quad \text{(electronic energy only)}$$
$$\Delta E_0 = \Delta E + \Delta E_{\text{ZPE}} \quad \text{(ZPE-corrected)}$$
$$\Delta H = \Delta E_0 + \Delta H_{\text{thermal}} \quad \text{(includes thermal corrections)}$$
$$\Delta G = \Delta H - T\Delta S \quad \text{(includes entropy)}$$

**Hierarchy of Accuracy**

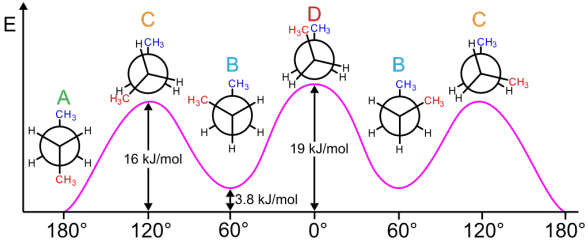
$$\Delta E \neq \Delta E_0 \neq \Delta H \neq \Delta G$$

You **must** include ZPE and thermal corrections for meaningful comparison with experiment!

**Equilibrium constant:**

$$K = \mathrm{e}^{-\Delta G^\circ / RT}$$

# Example: Conformational Energies of Butane




**DFT can predict:**

- Relative energies of conformers
- Rotational barriers
- Population ratios via Boltzmann distribution:

$$\frac{N_i}{N_j} = \frac{g_i}{g_j} \mathrm{e}^{-(E_i - E_j)/k_B T}$$

**At 298 K:**

$\Delta G(\text{gauche} - \text{anti})$
$\approx 0.9$ kcal/mol
$\Rightarrow$ ~82% anti, ~18% gauche

# Reading the Output: From Frequencies to $G$

**After a geometry optimization + frequency calculation**, the output file contains everything you need.

**Typical Gaussian output** (near the end of the log file):

```
Zero-point correction=              0.04489 (Hartree/Particle)
Thermal correction to Energy=       0.04773
Thermal correction to Enthalpy=     0.04867
Thermal correction to Gibbs Free Energy= 0.02236
Sum of electronic and zero-point Energies=    -76.38295
Sum of electronic and thermal Energies=       -76.38011
Sum of electronic and thermal Enthalpies=     -76.37917
Sum of electronic and thermal Free Energies=  -76.40548
```

**To compute $\Delta G^\circ$ for a reaction:**

1. Read "Sum of electronic and thermal Free Energies" for **each** species
2. $\Delta G^\circ = \sum G_{\text{products}} - \sum G_{\text{reactants}}$
3. $K = \mathrm{e}^{-\Delta G^\circ / RT}$

In ORCA: look for "Final Gibbs free enthalpy" or "$G$(298 K)" in the thermochemistry block.

# Limitations of the Harmonic Approximation

**The harmonic approximation breaks down for:**

1. **Large-amplitude motions:** internal rotations, ring puckering
   - Low frequencies ($< 100$ cm$^{-1}$) are most problematic
   - These contribute significantly to $S_{\text{vib}}$ and $G$
2. **Anharmonic vibrations:** bond dissociation, highly excited states
3. **Near-zero frequencies:** flat regions of PES (e.g., methyl rotations)

**Practical fixes:**

- **Quasi-harmonic correction** (Truhlar): raise all frequencies $< 100$ cm$^{-1}$ to 100 cm$^{-1}$
- **Quasi-RRHO** (Grimme): interpolate between HO and free-rotor models below a cutoff ($\sim$100 cm$^{-1}$)
- **ZPE scaling factors:** multiply calculated frequencies by an empirical factor (e.g., 0.9804 for B3LYP/6-31G(d))

# Summary of Lecture 1

1. **Hamiltonian → Born–Oppenheimer:** Separate electronic and nuclear problems
2. **Electronic structure:** HF treats exchange exactly but misses correlation; DFT includes both via $E_{\text{XC}}[\rho]$
3. **Kohn–Sham DFT:** Solve one-electron equations self-consistently; accuracy depends on the chosen functional
4. **PES:** Energy as a function of nuclear coordinates; minima and saddle points
5. **Harmonic approximation:** Taylor expansion of PES → force constants → Hessian → normal modes
6. **Partition functions:** $q = q_{\text{trans}} \cdot q_{\text{rot}} \cdot q_{\text{vib}} \cdot q_{\text{elec}}$, each computed from DFT output
7. **Thermodynamics:** $G = E_{\text{elec}} + E_{\text{ZPE}} + H_{\text{thermal}} - TS$

**Next lecture:** Transition states, reaction mechanisms, and energy profiles!

# Key Equations to Remember

**Electronic Structure:**

$$\hat{H}\Psi = E\Psi$$
$$E_{\mathsf{KS}} = T_s + E_{\mathsf{ext}} + J + E_{\mathsf{XC}}$$

**Harmonic Approximation:**

$$V(r) \approx \tfrac{1}{2}k(r - r_e)^2$$
$$\varepsilon_n = (n + \tfrac{1}{2})h\nu$$

**Partition Functions:**

$$q = q_{\mathsf{trans}} \cdot q_{\mathsf{rot}} \cdot q_{\mathsf{vib}} \cdot q_{\mathsf{elec}}$$
$$q_{\mathsf{vib}} = \prod_k \frac{1}{1 - \mathrm{e}^{-\Theta_{V,k}/T}}$$

**Thermodynamics:**

$$G = E_{\mathsf{elec}} + E_{\mathsf{ZPE}} + H_{\mathsf{th}} - TS$$
$$K = \mathrm{e}^{-\Delta G^\circ/RT}$$